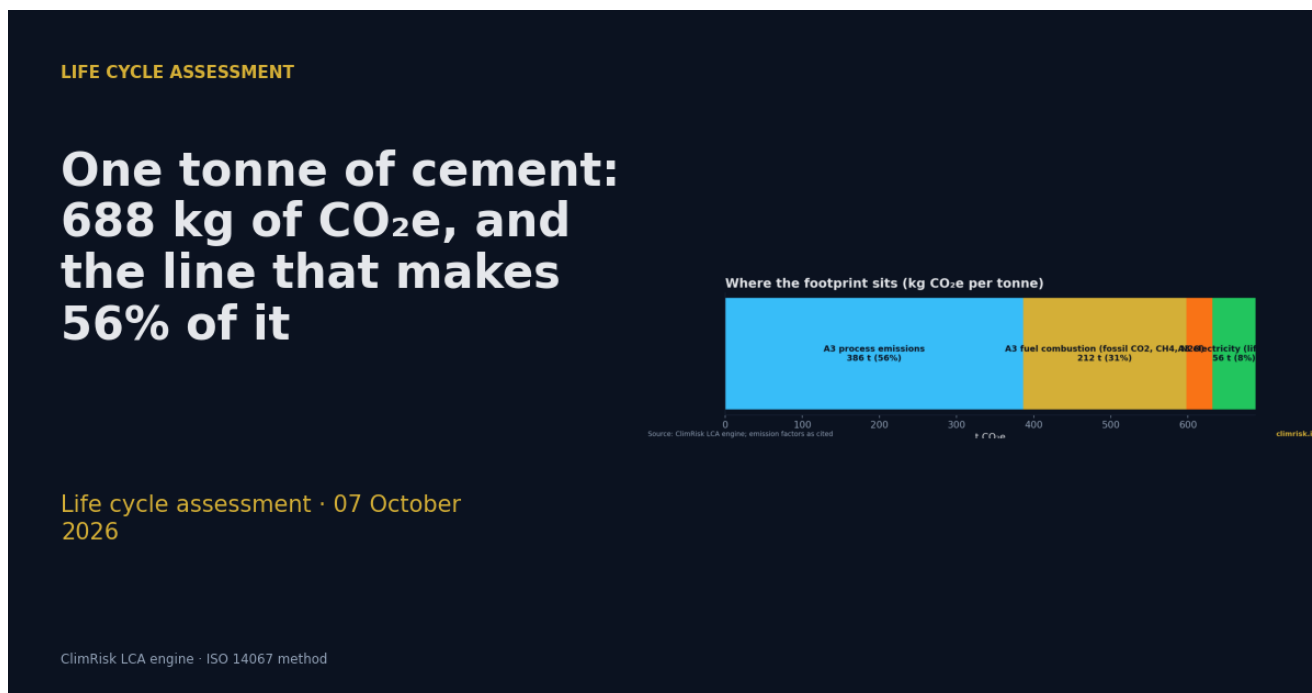


One tonne of cement: 688 kg of CO₂e, and the line that makes 56% of it

Cradle-to-gate footprint of 1 t cement made in IND: 688 kg CO₂e (5th–95th percentile 573–822). The largest single line is calcination & raw-meal carbon (gccca-csi default 525 kg co₂/t clinker (composition not provided)) (A3), at 386 kg.

ClimRisk LCA engine · 2026-10-07



2026. Every tonne of cement that leaves a gate carries a number, and almost nobody sees how it is made up. For this one, made in IND, the engine adds up 8 lines into 688 kg CO₂e. One line, calcination & raw-meal carbon (gccca-csi default 525 kg co₂/t clinker (composition not provided)), accounts for 56% of it [1].

One tonne of cement in six numbers

A cradle-to-gate footprint, taken apart

688 kg

CO₂e per tonne

ClimRisk LCA engine

573-822

5th-95th percentile, kg

Monte Carlo on factor uncertainty

56%

from calcination & raw-meal carbon

process chemistry

24.5 kg

biogenic CO₂, reported separately

ISO 14067 §6.3.5

poor

data quality, score 3.12

pedigree matrix

44%

primary-data share

of emissions

Source: ClimRisk LCA engine; factor sources as listed

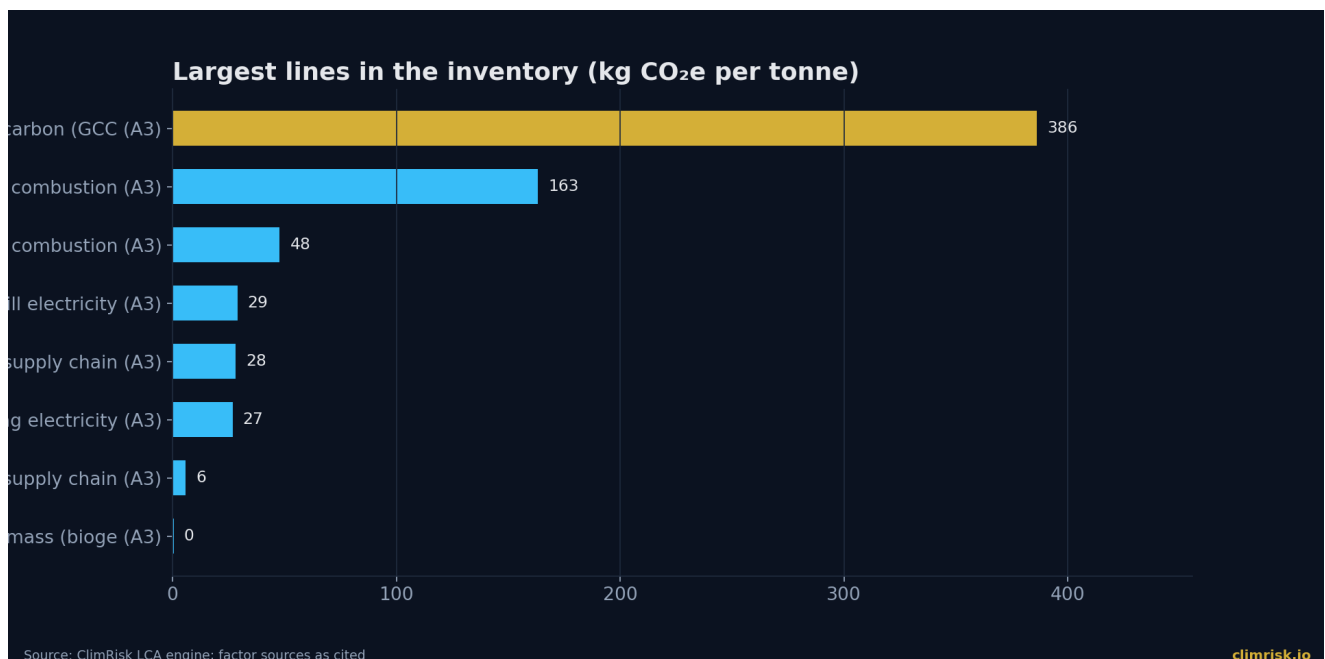
climrisk.io

Each number is computed from the inventory. Sources are stamped on the image.

Calcination & raw-meal carbon (GCCA-CSI default 525 kg CO₂/t clinker (composition not provided)): 56% of the footprint

The headline number is one figure; the decision is in the line that dominates it. Where the emissions sit decides which policies reach the site.

The loud number: where it comes from



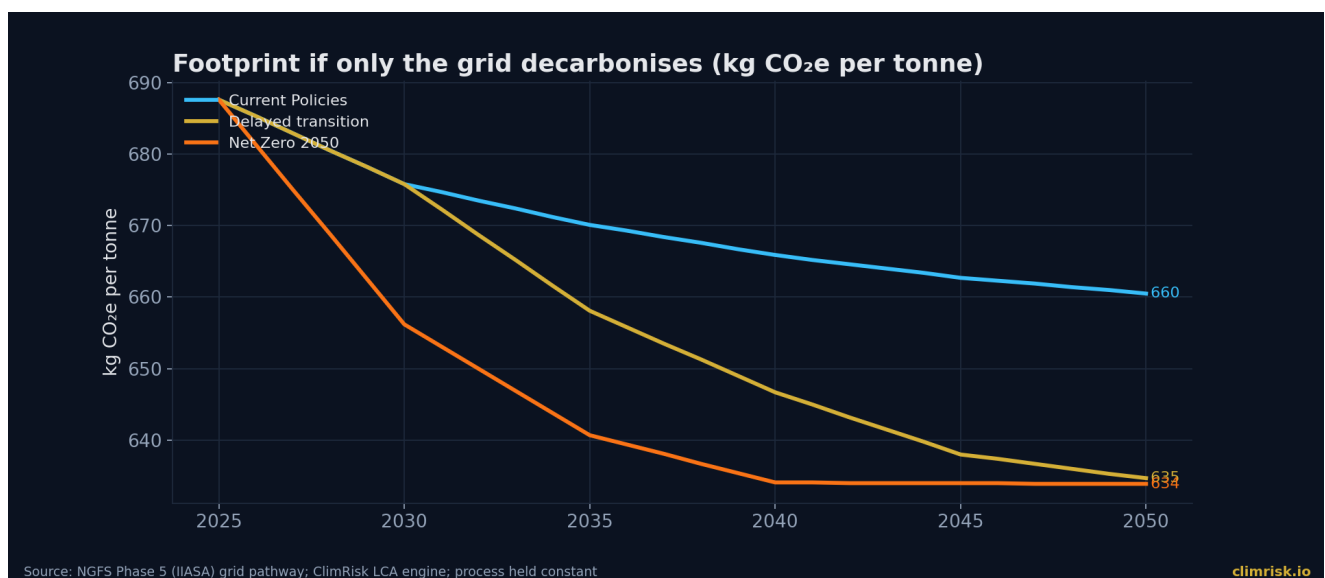
Largest contributions in the inventory; each line keeps its basis.

Stage	Item	kg CO ₂ e / t	Basis
A3	Calcination & raw-meal carbon (GCCA-CSI default 525 kg CO ₂ /t clinker (composition not provided))	386.2	process chemistry
A3	Other bituminous coal combustion	163.3	1.71 GJ × IPCC 2006 defaults
A3	Petroleum coke combustion	47.9	0.49 GJ × IPCC 2006 defaults
A3	Kiln & raw mill electricity	29.0	43.2 kWh × 670 g/kWh (Ember 2025 India (lifecycle))
A3	Other bituminous coal supply chain	28.2	DESNZ 2025 well-to-tank
A3	Cement grinding electricity	26.8	40 kWh × 670 g/kWh (Ember 2025 India (lifecycle))
A3	Petroleum coke supply chain	5.8	DESNZ 2025 well-to-tank
A3	Other primary solid biomass (biogenic) combustion	0.5	0.245 GJ × IPCC 2006 defaults (CO ₂ biogenic, reported separately)

The quiet part: what moves it, and what it hides

Clinker is 72% of the cement. Calcination alone releases 536 kg per tonne of clinker, and that is chemistry: GCCA-CSI default 525 kg CO₂/t clinker (composition not provided). Where the emissions sit decides which policies reach the site.

The uncertainty band is 18% either side of the median (lognormal per flow; CV by data quality (1:2%, 2:5%, 3:10%, 4:20%, 5:35%)) combined with factor uncertainty (IPCC CO₂ ±3%, CH₄/N₂O ±50%, WTT ±). Excluded: Other primary solid biomass (biogenic) supply chain (no well-to-tank factor available).



Grid-pathway effect only: the plant's own process is held constant, which is exactly why the line flattens.

What the border sees

Under EU CBAM the embedded emissions are 0.65 t per tonne (0.60 direct, 0.06 indirect). In 2026 the phase-in share is 2.5%, giving an indicative cost of \$1.39 per tonne at an EU ETS price of \$85/t [2]. Cement: CBAM counts direct and indirect emissions (Regulation (EU) 2023/956).

What this means, by reader

For the producer. The hotspot rows are the order of work; the biggest line (calcination & raw-meal carbon (gccca-csi default 525 kg co2/t clinker (composition not provided))) is rarely the one with the cheapest fix.

For the buyer. Ask for the 5th–95th percentile band and the data-quality rating with any single number.

Method and limitations

ISO 14067:2018 methodology, partial CFP cradle-to-gate (A1–A3); GWP100 IPCC AR6; biogenic CO2 reported separately. Run POST /lca/study for distribution, use stage, end of life, allocation, cut-off, sensitivity, data quality and the study report. Emission factors are cited per line [3]. Data quality is rated poor, with 44% of emissions from primary data. Calcination uses the GCCA-CSI default; provide clinker CaO/MgO for a Tier 2 result.

How ClimRisk measured this

- Inventory → fuels, electricity, process and transport flows per tonne, with factors from named sources.
- GWP → IPCC AR6 100-year values [4].
- Uncertainty → Monte Carlo on factor and activity-data ranges.
- Output → total, stage split, hotspots and the sensitivity of the total to each line.

Sources

1. ISO 14067:2018 Greenhouse gases — Carbon footprint of products <https://www.iso.org/standard/71206.html>
2. Regulation (EU) 2023/956 establishing a carbon border adjustment mechanism <https://eur-lex.europa.eu/eli/reg/2023/956/oj>
3. IPCC 2006 Guidelines Vol.2 (combustion) & Vol.3 Ch.2 (cement) — emission-factor source
4. IPCC AR6 WG1 Ch.7, GWP100 values <https://www.ipcc.ch/report/ar6/wg1/chapter/chapter-7/>

Produced by the ClimRisk engine from the sources listed. Model outputs with the limitations stated; not forecasts or investment advice.